

How Did We Get From This...

1950S+ RULE-BASED

```
IF temperature > 38°C  
THEN fever
```

2020S+ GENERATIVE

Write a space poem...

Deep in silicon neurons...

...to ChatGPT?

The answer lies in understanding four important concepts:



Artificial Intelligence



Machine Learning



Deep Learning



Generative AI

Swipe to travel through time →

Let's Travel Through Time

AI didn't appear overnight. The AI we use today is the result of decades of innovation. Every stage solved a problem the previous stage couldn't.



Rule-Based Systems

Decisions programmed by hand

1950s



Machine Learning

Learning from examples & data

1980s



Deep Learning

Multilayer Neural Networks

2010s



Generative AI

Creating brand-new content

2020s

Let's see how we got here.

Stage 1: Artificial Intelligence

The Goal: Build machines that can perform tasks requiring human intelligence.

- ✓ Solve problems
- ✓ Understand info

- ✓ Make decisions
- ✓ Assist humans

TRIGGER INPUT

 It starts raining

RULE MATCHED

IF raining → carry umbrella

ACTION

Carry Umbrella 

 Chess programs

 Expert systems

 Rule-based logic

 RED Light → Stop

Thousands of rules. Millions of possibilities.

 LIMITATION

The Problem With Rule-Based AI

Rule-based systems worked well... until the world became too complex. Writing rules manually for everything became impossible.



The rules change?

Requires manually editing thousands of lines of hardcoded code.



New situations appear?

The system breaks because it cannot adapt to unprogrammed cases.



Millions of possibilities exist?

Manually scripting millions of decision branches is completely impossible.

- **SYSTEM HALTED: COMPLEXITY_LIMIT_EXCEEDED**

Unstructured input encountered. Real-world logic complexity bounds exceeded.

We needed machines that could learn instead of simply obey.

Stage 2: Machine Learning

Instead of teaching rules, teach using **examples**. Show the machine thousands of emails and it learns to classify them.



EXAMPLES OF MACHINE LEARNING

 Spam detection

 Fraud detection

 Netflix recommends

 Amazon suggestions

 Demand forecasting

The Problem With Traditional ML

Machine Learning was a huge breakthrough. But complex, high-dimensional tasks were still extremely difficult.



Visual Perception

Is this a cat or dog? Traditional ML gets lost in millions of raw pixel combinations.



Audio/Speech

What did they say? High complexity due to speed, accent variations, and noise.



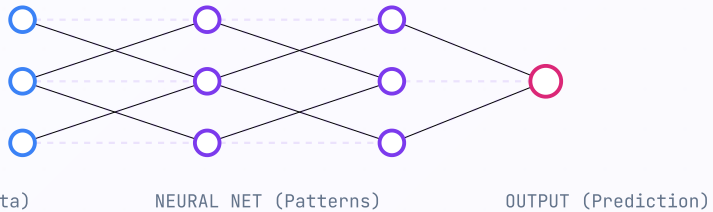
Self-Driving Scenarios

Identifying pedestrians, road signs, and vehicles dynamically in real-time requires deep pattern understanding.

Scientists needed something much more powerful.

Stage 3: Deep Learning

Researchers built multi-layered **Neural Networks** capable of learning highly complex patterns directly from raw data.



EXAMPLES

- 📱 Face Unlock & Vision
- 🗣️ Voice Assistants
- 🌐 Language Translation
- 🚗 Self-driving systems

MACHINES LEARNED TO

- 👁️ See patterns
- 🎤 Hear speech
- 🧠 Understand language

Then Came A New Question

AI could now categorize, classify, and recognize patterns with high precision. But what if it could go a step further and **create**?

TRADITIONAL AI

- ✓ Classify data
- ✓ Predict values
- ✓ Recognize items



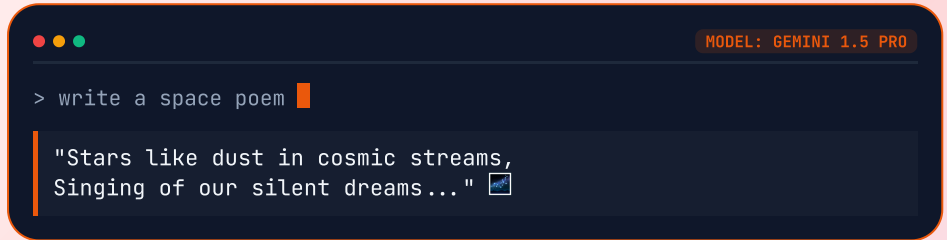
GENERATIVE AI

- ✍ Write articles
- 🖌 Generate images
- 🎵 Compose music
- 📄 Write code

This question led to the next revolution.

Stage 4: Generative AI

Generative AI creates **brand-new content**: text 🖋️, images 🎨, music 🎵, videos 🎬, and code 💻.



ChatGPT



Claude



Gemini



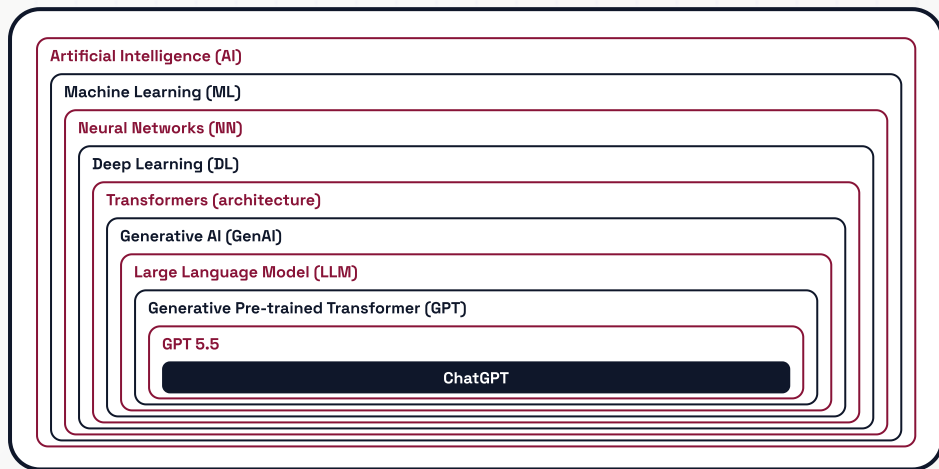
Midjourney



Sora

This is the technology that captured the world's attention.

The Evolution Of AI



Decades of innovation built layer by layer:

- **AI:** Follow instructions
- **ML:** Learn from data
- **DL:** Complex patterns
- **GenAI:** Create brand-new content

 **Follow Resnur AI**



One concept. Every single day.